

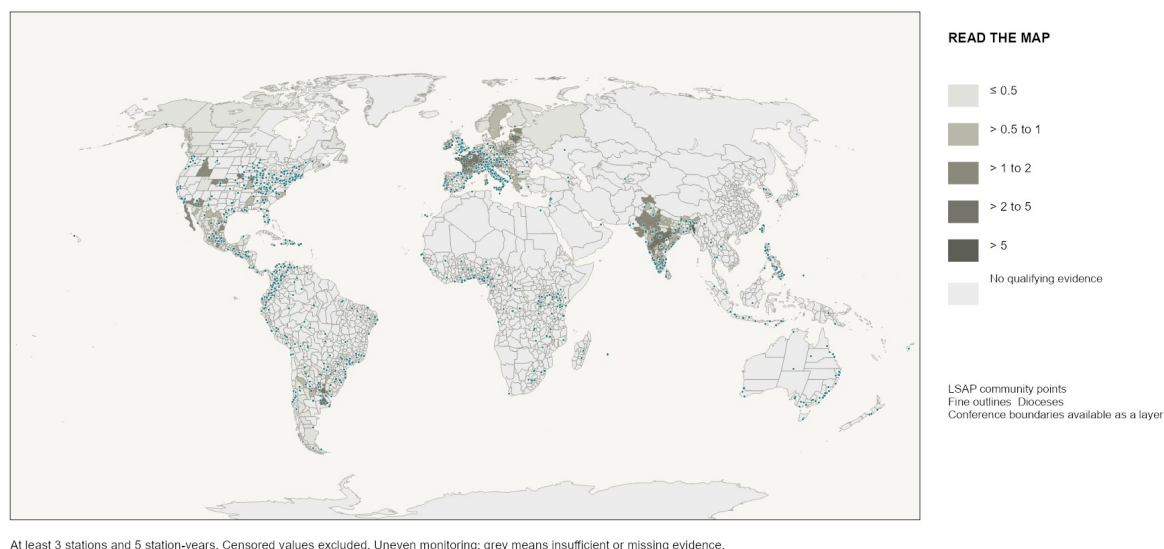
Nitrate in monitored diocesan rivers

This report explains the source data, executed workflow and interpretation of nitrate in monitored diocesan rivers. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Heinle et al., 2026).

Map P03 | Documentation date 6 October 2026 | Units mg/L as nitrogen | River observation window 2015–2024

Nitrate in monitored diocesan rivers

Median of river station year medians · 2015–2024 · mg/L as N



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LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P03

Figure P03 World overview exported from the completed ArcGIS Pro layout.

At least 3 stations and 5 station-years. Censored values excluded. Uneven monitoring; grey means insufficient or missing evidence.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Heinle et al., 2026).

1. Download GFQA v3 to a dedicated temporary FreshwaterScratch folder. Read station and measurement records from the archive in memory; create no persistent station layer.
2. Select river stations, observations from 2015–2024, the exact parameter and mg/L units. Exclude flagged censored values, bad/poor/suspect quality, negatives and nonfinite results.
3. Assign stations to copied dioceses using point-in-polygon; deterministic smallest DIO_ID resolves overlapping/boundary matches.
4. Calculate each station-year median, then the diocesan median and interquartile range of station-year medians. Retain counts of distinct stations, station-years and monitored years.
5. Publish only when at least three stations and five station-years qualify. Suppress derived numeric values for smaller groups; retain all diocesan polygons and narrative coverage status.
6. Symbolize Median using boundaries 0.5, 1, 2 and 5 mg/L as N. Reverse the palette for dissolved oxygen so lower oxygen is darker. Create diocesan story popups.
7. Verify only aggregated freshwater feature classes are referenced by the project; remove the downloaded archive and all scratch freshwater source records. Retain aggregate QA, source attribution and reproducible scripts.

Output fields and files

Field or band	Meaning
Median	Median of qualifying station year medians
IQR	Interquartile range of station year medians
Stations and StationYears	Distinct stations and station year count
Years	Distinct monitored years in selected window
Coverage	Reporting status rather than area coverage
WaterStory	Monitoring and interpretation narrative
Parameter Units Period	Parameter identity, mg/L and 2015–2024

Analysis dataset: PollutionAnalysis.gdb/Freshwater_NO3N. Styles/P03.lyrx and Styles/P03.mapx preserve layer and map definitions. Map_Exports/P03_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

630 dioceses meet reporting thresholds; 357,971 measurements were accepted. All 2,874 copied dioceses are retained; missing and suppressed summaries remain null (LSAP Pollution Atlas, 2026, freshwater_aggregate_QA.json).

Interpretation and data handling

These statistics describe monitored rivers, not every waterbody. Censored-value exclusion can bias concentrations upward. Station-year medians give each station-year equal weight; they are not area-weighted water surfaces. GFQA v3, CC BY 4.0. Raw-data removal follows the user’s output requirement; deletion alone does not establish compliance with every possible use condition.

FreshwaterScratch/GFQA_v3.zip and the extracted method metadata were deleted after validating the aggregate outputs and project connections. No raw station observations or station geometry layer remains in this pollution project. Aggregate QA, nonobservational source metadata and scripts are retained. CC BY 4.0 attribution remains required; source removal was an output requirement, not a claim that deletion alone determines licence compliance (Heinle et al., 2026; LSAP Pollution Atlas, 2026, cleanup_audit.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Heinle, M., Lisniak, D., & Saile, P. (2026). UNEP GEMS/Water Global Freshwater Quality Archive (Version v3) [Data set]. Zenodo. Published 2 February 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.